BETHANY HIGH SCHOOL, Sarjapur, Bangalore
FIRST TERM RIVISION WORKSHEET
Std – VIII
CHEMISTRY

SECTION-A

Question 1:

Choose the correct answers to the questions from the given options:

(i) Which of the following is NOT a sign of a chemical reaction?
(a) Formation of gas
(b) Change in colour
(c) Change in temperature
(d) Change in shape

(ii) Consider the following statements regarding chemical reactions and catalysis:
I. A promoter is a substance that enhances the activity of a catalyst but does not usually act as a catalyst by itself.
II. Thermal dissociation is a chemical change in which a compound breaks down into simpler substances on heating.
III. In a catalytic reaction, a catalyst alters the rate of the reaction without being permanently consumed.
Which of the above statements is/are correct?
(a) I only
(b) II only
(c) I and III only
(d) I, II and III

(iii) **Assertion:** Atoms can combine either by transfer of valance electrons from one atom to another or by sharing of valence electrons.
Reason: Sharing and transfer of valence electrons is done by atoms to have an octet or duplet in their valence shell.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

(iv) Which of the following is an example of a thermal decomposition reaction?
(a) $2H_2 + O_2 \rightarrow 2H_2O$
(b) $2KClO_3 \rightarrow 2KCl + 3O_2$
(c) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
(d) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

(v) An atom X has an atomic number of 12 and mass number of 24. The number of electrons, protons, and neutrons it contains are respectively:
(a) 12, 12, 24
(b) 12, 12, 12
(c) 12, 24, 12
(d) 24, 12, 12

(vi) Which property allows hydrogen gas to be collected by the downward displacement of water?
(a) It is highly soluble in water and lighter than air.
(b) It is practically insoluble in water and lighter than air.
(c) It is heavier than air.
(d) It reacts vigorously with water.

(vii) $Ba\ (s) + F_2\ (g) \rightarrow BaF_2\ (s)$
The above reaction is an example of:
(a) combination
(b) double decomposition
(c) decomposition
(d) displacement

(viii) During the formation of a covalent molecule like Carbon Tetrachloride, how many pairs of electrons are shared between carbon and the chlorine atoms in total?
(a) 1
(b) 2
(c) 4
(d) 8

(ix) In the Bosch Process for the industrial manufacture of hydrogen, the catalyst used is:
(a) Finely divided Iron
(b) Ferrous oxide
(c) Chromic oxide
(d) Ferric oxide

(x) Total number of electrons that take part in forming bonds in N_2 is:
(a) 2
(b) 4
(c) 6
(d) 8

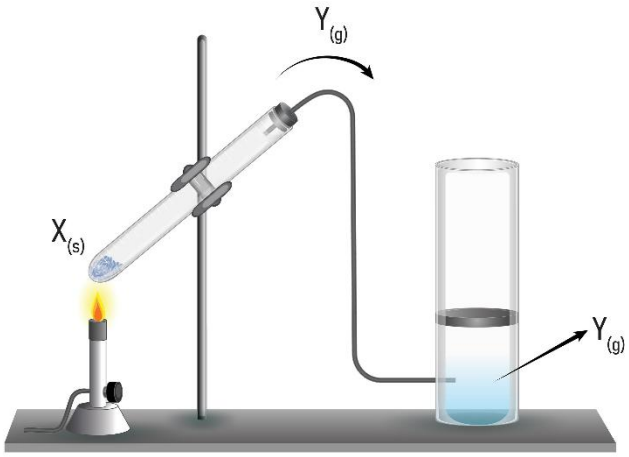
Question 2:

(i) Choose the correct option from the brackets to complete each statement.
(a) When hydrogen gas is reacts with heated copper (II) oxide, it acts as a/an _____ agent and forms water and copper. [reducing / oxidising]
(b) The electro-valency of magnesium in magnesium chloride ($MgCl_2$) is _____. [+2 / +1]
(c) During the laboratory preparation of hydrogen using granulated zinc and dilute hydrochloric acid, _____ is added as a catalyst to accelerate the reaction. [copper sulphate / sodium chloride]
(d) An atom or a group of atoms carrying a negative charge is called an _____. [anion / cation]
(e) The reaction between an acid and a base to form salt and water is known as a _____ reaction. [neutralisation / decomposition]

(ii) Identify the Scientific Term:
(a) A reaction in which two or more substances combine to form a single substance.
(b) The subatomic particle whose numbers vary in the isotopes of an element.
(c) The chemical bond formed due to the electrostatic force of attraction between oppositely charged ions.
(d) The technique used to collect hydrogen gas in the laboratory because the gas is practically insoluble in water.

(e) A pair of electrons in the valence shell of an atom that does not take part in bond formation or sharing with another atom.

(iii) The test tube contains a blue solid Compound X. When heated strongly, the blue solid changes into a black residue Compound Z, and a reddish-brown gas Y and oxygen escapes through the delivery tube into a gas jar.



$$X_{(s)} + \text{heat} \longrightarrow Y_{(g)} + Z_{(s)} + O_2$$

(a) Identify the blue chemical compound X and the black residue Z.

(b) Name the Y which produced along with oxygen.

(c) Write a fully balanced chemical equation for this reaction.

(d) Identify the type of chemical reaction shown.

(e) Is it a reversible or an irreversible chemical change? Justify your answer.

SECTION-B

Question 3:

(i) Define:

(a) Octet and duplet rules

(b) Chemical bonding

(c) Exothermic reaction

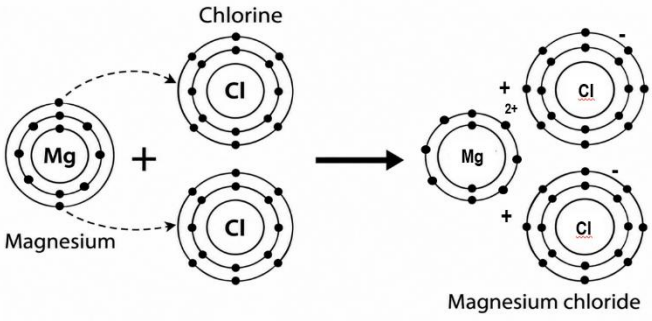
(ii) Difference between:

(a) Stable and unstable electronic configuration

(b) Electrovalent and covalent bond

(c) Mass number and atomic number

(iii) Look at the electronic transfer diagram below representing the formation of Magnesium Chloride and answer the questions that follow:



(a) Write the total number of valence electrons present in a neutral Magnesium atom.

(b) State the nature of the chemical bond formed here.

(c) Why does a magnesium atom readily lose electrons to chlorine atoms instead of gaining them? Explain in terms of the Octet Rule.

Question 4:

(i) Answer the following:

In the atom of an element 'Z', 5 electrons are present in the outermost shell. It requires a noble gas configuration by accepting a requisite number of electrons.

(a) What would be the charge on the ion so formed?

(b) Write the formula of the compound which will be formed when 'Z' reacts with Na atom.

(ii) The atomic number of lithium is 3. Its mass number is 7.

(a) How many protons and neutrons are present in a lithium atom?

(b) Draw the diagram of a lithium atom.

(iii) Give reasons:

(a) Silver nitrate solution is kept in dark coloured bottles.

(b) Finely divided iron used in the manufacture of ammonia.

(c) Blue solution of copper sulphate changes to green when a piece of iron is added to this solution.

(iv) What do you observe when?

(a) Lead nitrate is heated.

(b) Silver chloride is exposed to sunlight.

(c) Sodium chloride solution is added to silver nitrate solution.

Question 5:

(i) An element 'X' contains 6 electrons in 'M' shell as valence electrons:

(a) What is the atomic number of 'X'?

(b) Identify whether 'X' is a metal or non-metal.

(ii) An element 'Z' forms the following compound when it reacts with hydrogen, chlorine, oxygen. ZH_3, ZCl_3 and Z_2O_3

(a) What is the valency of element 'Z'?

(b) Element 'Z' is metal or non-metal?

(iii) Write down the electron distribution of chlorine atoms. How many electrons are there in the L shell?

(iv) Name the following:

(a) The atom of an element has one electron, one proton and no neutron.

(b) An atom has no valency.

(c) Molecule has triple covalent bonds.

(d) An atom attaining overall negative charge.

Question 6:

(i) Write balanced chemical equations for the reactions where the following are involved.

(a) Heat

(b) Pressure

(c) Catalyst

(ii) Study the incomplete table given below regarding three different neutral atoms (P, Q, and R) and answer the questions that follow:

Atom	Atomic Number	Mass Number	Number of Protons	Number of Neutrons	Electronic Configuration
P	17	35	A	18	2,8,7
Q	B	24	12	12	C
R	17	37	17	D	2,8,7

(a) Fill in the blank spaces A, B, C and B in the table.

(b) Identify which two atoms represent a pair of isotopes.

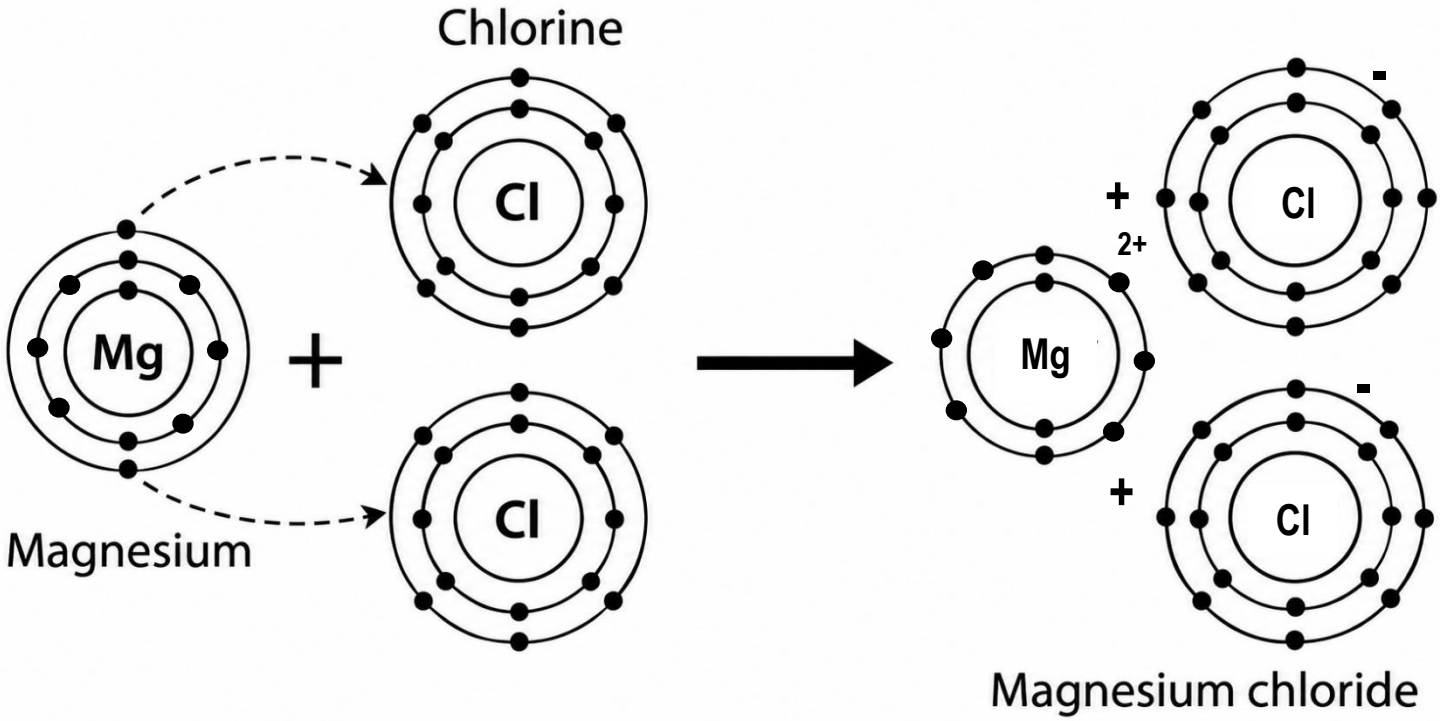
(c) State the total number of valence electrons present in atom Q.

(iii) Write the chemical equation for following:

(a) Silver oxide undergoes thermal decomposition

(b) Mercury (II) oxide is heated

(c) Hydrated copper sulphate is heated


Chlorine
Cl
Mg
+
Cl
Magnesium
2+
+
Cl
-
Mg
+
Cl
-
Magnesium chloride